Editor

1exp1.m | MaheshCo1ex2.m | Maheshco1exp3.m | MaheshCo1exp4.m

/MATLAB Drive/MaheshCo1exp4.m

```
clc;
clear;
close all;

%% 1. INPUT PARAMETERS
Fs = 1000;
Fp = 100;
Fst = 150;
Rp = 1;
Rs = 40;

Wp = Fp/(Fs/2);
Wst = Fst/(Fs/2);

%% 2. CHEBYSHEV TYPE-I THEORETICAL ORDER
epsilon = sqrt(10^(Rp/10)-1);

% Bilinear-transform prewarped frequency ratio
Omega_p = tan(pi*Fp/Fs);
Omega_s = tan(pi*Fst/Fs);

N_formula = acosh(sqrt((10^(Rs/10)-1)/(10^(Rp/10)-1))) ...
            / acosh(Omega_s/Omega_p);

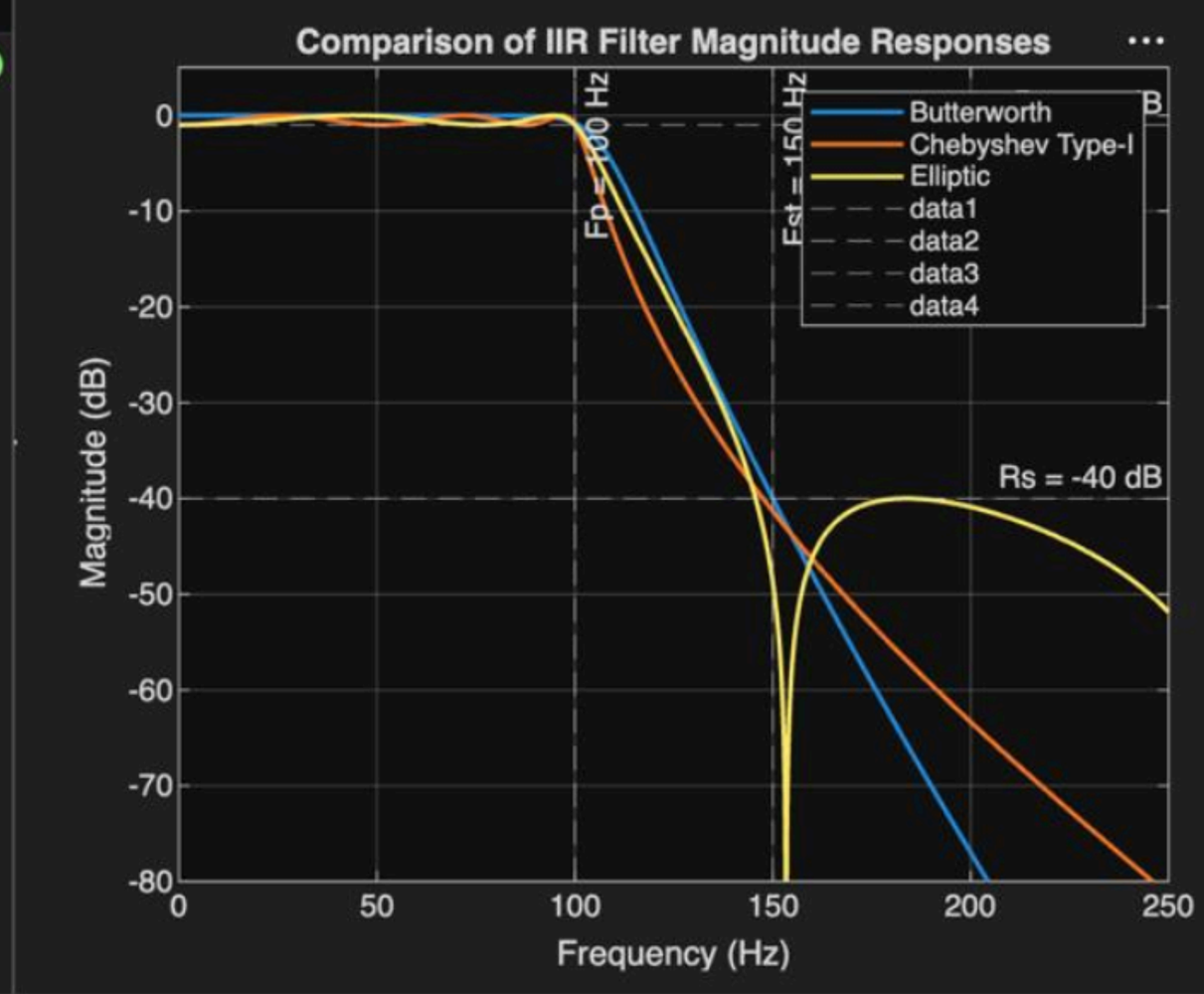
N_theory = ceil(N_formula);

%% 3. BUTTERWORTH FILTER
[Nb,Wb] = buttord(Wp,Wst,Rp,Rs);
[bB,aB] = butter(Nb,Wb,'low');

%% 4. CHEBYSHEV TYPE-I FILTER
[Nc,Wc] = cheb1ord(Wp,Wst,Rp,Rs);
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Figures

Figure 1 | Figure 2 | Figure 3



Command Window

New to MATLAB? See resources for Getting Started.

```
Chebyshev Type-I = 3.3413 %
Elliptic = 0.0000 %

TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Bandwidth = 50.00 Hz
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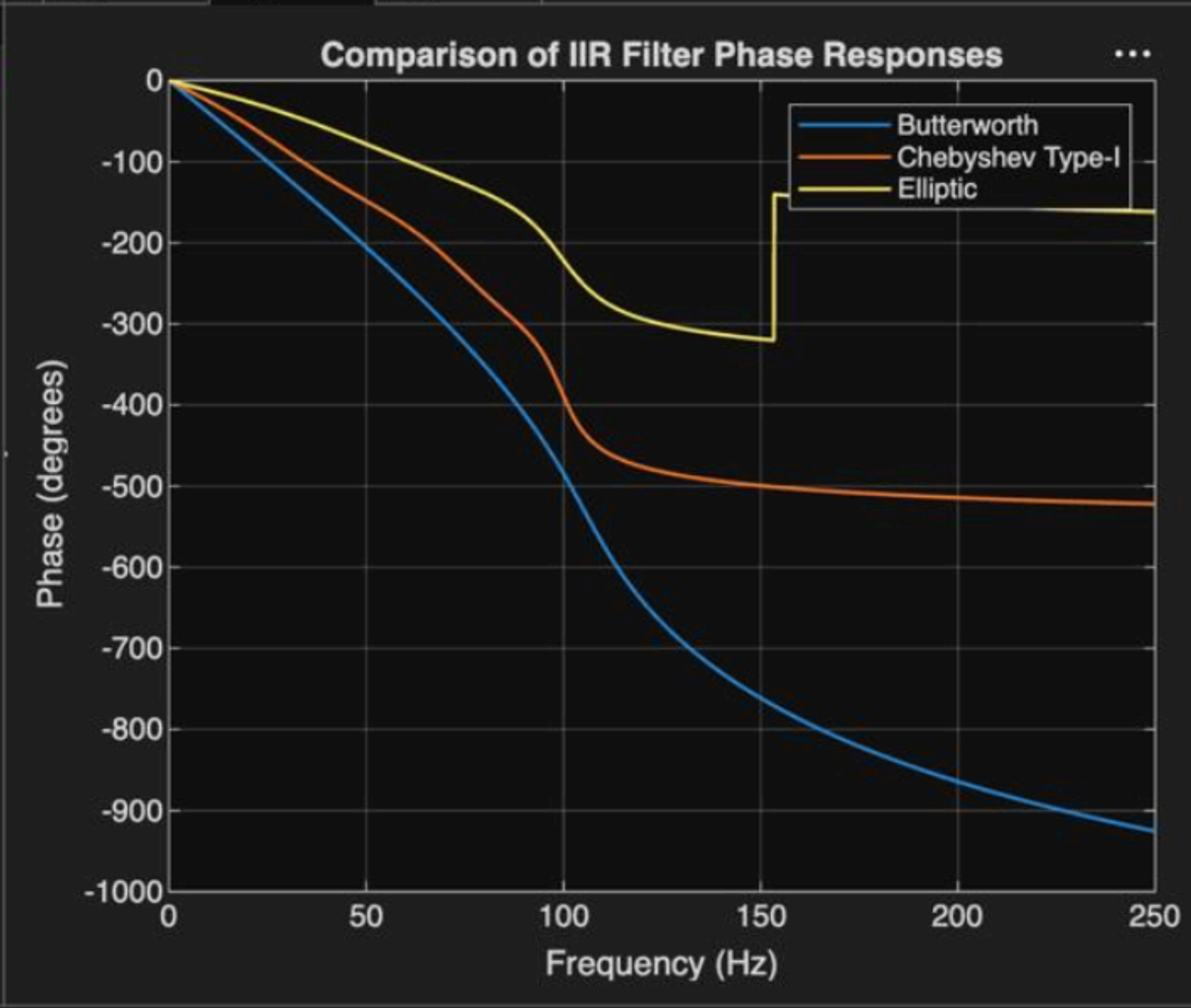
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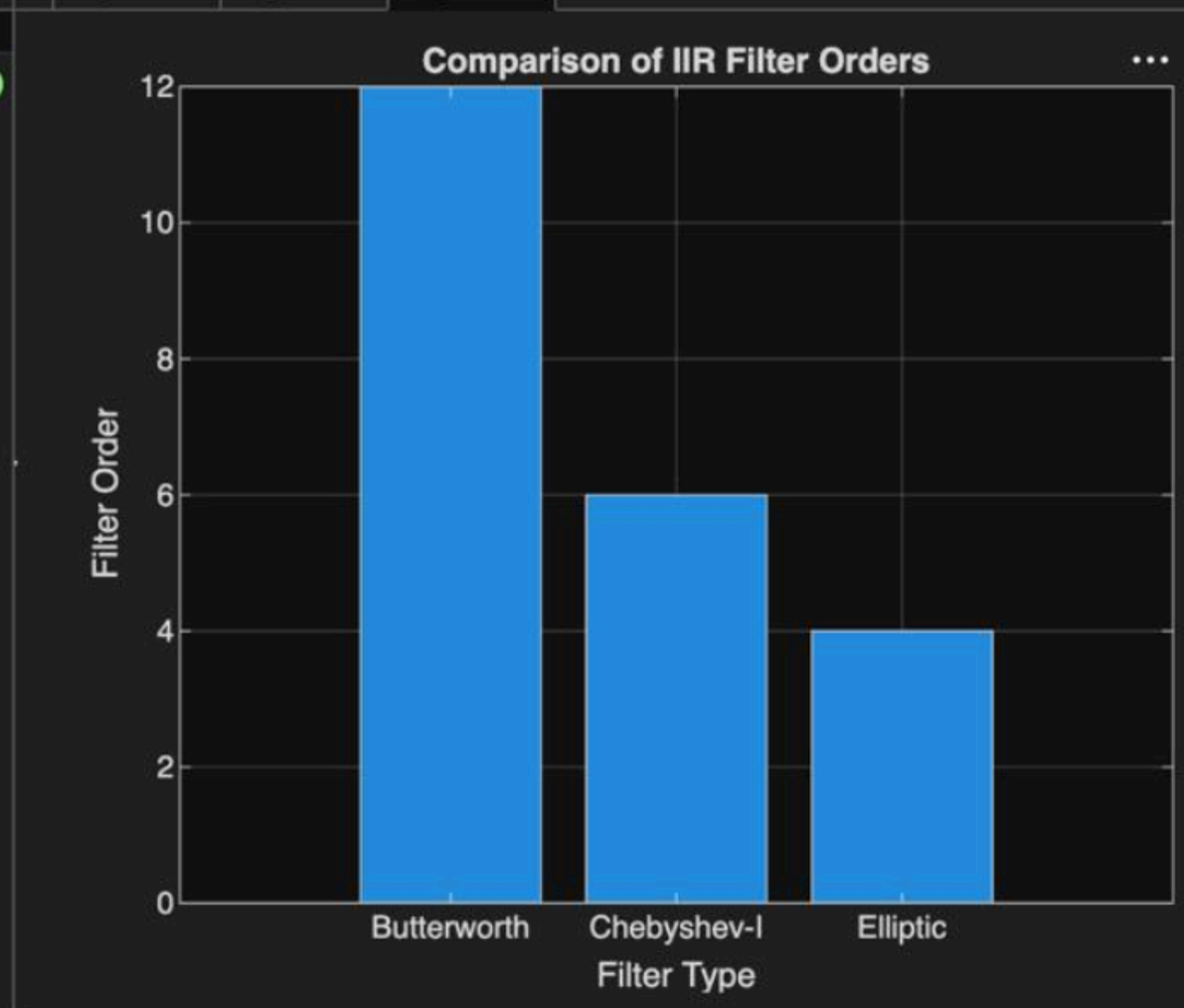
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